
Closing the Algebra Is Not Enough: Lie-Closure Regularization Makes Learned Transformation Structure Comparable, Not Identifiable

Anonymous Author(s)

Affiliation

Address

email

Abstract

When do independently trained models learn the same *transformation structure*, and when does agreement between them reflect the truth? We study this for transition-based representation learners that infer latent actions from pairs of observations and represent them with learned skew-symmetric generators. We add a Lie-closure penalty (BracketNet) and evaluate on $SO(2)$, T^2 and $SO(3)$ actions observed through nonlinear maps and through rendered images, with every choice fixed on development seeds and fresh seeds evaluated once.

(1) Transition fitting alone recovers the abelian T^2 algebra in most runs, but leaves the $SO(3)$ generator span non-closed in 15/20 runs. Closure regularization closes it, roughly halves the distance to the true algebra, and makes the algebras of independently trained models agree more (Holm-corrected $p \leq 8.8 \times 10^{-3}$), with no detectable change in transport error. (2) Closed is not correct. 8/20 $SO(3)$ runs close onto a *chiral* $\mathfrak{su}(2)$, and on rendered images every closed run (7/20) is chiral. We prove why the objective favours it: a rotation about the axis through a point leaves that point fixed, so the true action cannot be inferred from endpoints, whereas a chiral algebra acts simply transitively and makes transport and composition exact. Composition errors, a pre-registered intervention with true action coefficients, and a generator-span trace agree with this account. (3) Transfer of transformations between models tracks whether they learned the *same* algebra, even a wrong one. Linear CKA catches encoder failures but not these class mismatches.

Algebraic closure turns continuous disagreement between models into disagreement among a few closed subalgebras: it makes learned transformation structure comparable, but not identifiable.

1 Introduction

Representation-similarity methods ask whether two networks embed data in the same way, up to a class of transformations [Kornblith et al., 2019, Williams et al., 2021, Moschella et al., 2023]. For models that learn how data *transform*, there is a second question: do they agree on the transformations themselves? This matters when actions inferred by one model are to be used in another, and when a learned symmetry is read as a scientific statement. Embedding agreement does not settle it. Two

models can embed the same data almost identically and still represent its transformations by different Lie algebras, and two models can agree perfectly with each other while both being wrong (Cor. 1).

We study this in transition-based symmetry learning [Park et al., 2022, Keurti et al., 2023, Yang et al., 2024]. A model infers a latent action from a pair of observations and represents it through learned generators. Nothing in a transition loss requires those generators to span a Lie algebra, so a natural structural bias is to penalize commutators that leave their span. We ask: *when do independently learned transformation representations become structurally comparable, and why does algebraic consistency fail to make them identifiable?*

Contributions.

- **An evaluation that separates five levels of agreement:** embedding, algebra, representation class, inferred action and ground truth. It includes cross-model transformation transfer, uses disjoint seed pairs as independent units, and pre-registers its primary tests.
- **Evidence that closure changes the *kind* of disagreement.** For $SO(3)$, transition fitting leaves generator spans non-closed and scattered. A closure penalty concentrates them on the three closed classes allowed in $\mathfrak{so}(4)$, one correct and two chiral, which improves cross-model agreement without guaranteeing correctness.
- **A mechanism with a proof** (Prop. 3). Endpoint-based action inference cannot capture rotations about the axis through a point, whereas chiral algebras make transport and composition exact. The objective therefore favours a wrong closed algebra. Three tests, two of them pre-registered, support this account.
- **A transfer analysis:** actions transfer between models that learned the same algebra, right or wrong. CKA catches representation failures but not algebra-class mismatches.

We also give a corrected invariance analysis of the closure loss and a local bound on how far products of group elements leave the learned span. The underlying Lie theory is standard; we cite it as such.

2 Related work

Learning transformation groups. Lie-group transformation models have a long history [Rao and Ruderman, 1999, Sohl-Dickstein et al., 2010, Cohen and Welling, 2014]. Recent methods discover symmetries from predictors [Dehmamy et al., 2021], adversarially from distributional invariance [Yang et al., 2023], in learned latent spaces [Yang et al., 2024], beyond affine families [Shaw et al., 2024], or as one-parameter families [Gabel et al., 2023]. Equivariant networks assume a known group [Cohen and Welling, 2016, Bronstein et al., 2021].

Group-structured representations from transitions. Symmetric embedding networks [Park et al., 2022] use known actions. Homomorphism autoencoders [Keurti et al., 2023] learn a homomorphism from action labels into a matrix group without knowing the group. Symmetry-based disentanglement [Higgins et al., 2018] defines representations through group actions. We share the transition setting but use *no* action labels. Our oracle-coefficient intervention (Sec. 6.2) is the action-supervised counterpart and shows what the missing labels cost. LieGAN and LaLiGAN learn from distributional signals rather than transitions, so we do not compare against them directly.

Comparing representations. CKA [Kornblith et al., 2019], shape metrics [Williams et al., 2021] and relative representations [Moschella et al., 2023] compare embeddings up to a transformation class, and convergence of representations across models is an active hypothesis [Huh et al., 2024]. These tools compare *point clouds*. We compare the transformation algebras acting on them, and the transfer of actions between models.

What is not new. Closure of a matrix span, the classification of subalgebras of $\mathfrak{so}(4)$ and $\mathfrak{so}(5)$, Dynkin’s formula, the hairy-ball theorem and quaternion transitivity are standard. Our contribution

is their consequences for this learning problem and the empirical analysis. Appendix D compares settings in detail.

3 Setting and method

Observations $x = h(u)$ come from states u through an unknown injective map. Training data are unlabeled triples $(x_0, x_1, x_2) = (h(u), h(g_1 u), h(g_2 g_1 u))$ with near-identity g_i from an unknown compact group, and K is given.

The model has an encoder E , a decoder D , an action-inference network $a = q(z, z')$ (bounded by $0.65 \tanh$) and generators $A_k \in \mathfrak{so}(d)$, each rescaled to $\|A_k\|_F = \sqrt{2}$. An action is $\rho(a) = \exp(\sum_k a_k A_k)$.

With $z_t = E(x_t)$ and $a_{st} = q(z_s, z_t)$, the losses are:

- reconstruction $\sum_t \|D(z_t) - x_t\|^2$;
- transport in latent and observation space, $\sum_{t < 2} \|\rho(a_{t,t+1})z_t - z_{t+1}\|^2$ and $\sum_{t < 2} \|D(\rho(a_{t,t+1})z_t) - x_{t+1}\|^2$;
- composition $\|\rho(a_{02}) - \rho(a_{12})\rho(a_{01})\|_F^2$;
- a Gram penalty $\|B^\top B - 2I\|_F^2$ and an anti-collapse term $\|\text{Cov}(z) - I\|_F^2$;
- the closure residual

$$\mathcal{L}_{\text{close}} = \frac{1}{K^2} \sum_{i,j} \|(I - P_\varepsilon) \text{vec}[A_i, A_j]\|_2^2, \quad P_\varepsilon = B(B^\top B + \varepsilon I)^{-1} B^\top, \quad (1)$$

where B stacks the vectorized generators.

LOCAL uses reconstruction, transport and the regularizers. +COMP. adds composition (weight 1.2). BRACKETNET adds $\lambda(t)\mathcal{L}_{\text{close}}$: $\lambda(t) = 0$ for 35% of updates, a linear ramp to λ by 70%, and frozen generators afterwards. All three share networks, data and optimizer (weights in Appendix B).

4 Theory: what closure guarantees, and why it can prefer the wrong algebra

Notation: $\mathfrak{a} = \text{span}(A)$, $G = B^\top B$, P_0 the orthogonal projector onto $\mathfrak{a}$, $N = I - P_0$. Proofs are in Appendix A. Standard Lie-theoretic facts are cited as such; our contributions are their consequences for this learning objective.

Invariances. With $\varepsilon=0$ and $M \in GL(K)$, $\mathcal{L}_{\text{close}}(BM) = K^{-2} \|R(M \otimes M)\|_F^2$, where R is the residual matrix, so $\sigma_{\min}(M)^4 \mathcal{L}_{\text{close}} \leq \mathcal{L}_{\text{close}}(BM) \leq \sigma_{\max}(M)^4 \mathcal{L}_{\text{close}}$. The loss is invariant only under orthogonal reparameterizations; its zero set (“ $\mathfrak{a}$ is a subalgebra”) is $GL(K)$ -invariant. For $\varepsilon > 0$, $P_\varepsilon(BM) = B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$. Because $\text{diag } G = 2$, the Gram target affects no gradient.

Proposition 1 (Closure, escape and integration). (a) If $A_1, \dots, A_K$ are independent and $[A_i, A_j] \in \mathfrak{a}$, then $\mathfrak{a}$ is a Lie subalgebra and integrates to a connected immersed subgroup of $SO(d)$ (standard). (b) The uniform escape constant $\delta = \sup_{X,Y \in \mathfrak{a}} \|N[X, Y]\|_F / (\|X\|_F \|Y\|_F)$ satisfies $\delta \leq K \sqrt{\mathcal{L}_{\text{close}}(A; 0)} / \lambda_{\min}(G)$. (c) For $X, Y \in \mathfrak{a}$ with $\|X\|_F, \|Y\|_F \leq r < \frac{\ln 2}{4}$, $\text{dist}(\log(e^X e^Y), \mathfrak{a}) \leq \frac{\delta}{4} [-\ln(2 - e^{4r}) - 4r] = \frac{1}{2} \delta r^2 + O(\delta r^3)$. (d) Chained orthogonal transport accumulates per-step errors at most additively.

Part (c) is local: it covers rotations of up to ≈ 0.12 rad, smaller than typical inferred actions.

Closed subalgebras in the benchmarks. Write $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (two commuting ideals).

Proposition 2 (Classification; standard facts applied to our benchmarks). (a) Every 2-dimensional subalgebra of $\mathfrak{so}(5)$ is abelian and $SO(5)$ -conjugate to the true $\mathfrak{t}^2$. (b) Every 3-dimensional subalgebra of $\mathfrak{so}(4)$ is isomorphic to $\mathfrak{su}(2)$. It is $SO(4)$ -conjugate to exactly one of three: the diagonal

115 $\mathfrak{so}(3)$ acting on $\mathbb{R}^3 \oplus \mathbb{R}$ (the truth), $\mathfrak{su}(2)_L$ or $\mathfrak{su}(2)_R$. Under $O(4)$ the two chiral factors merge into
 116 one class. (c) No closed 4-dimensional span in $\mathfrak{so}(4)$ contains the diagonal $\mathfrak{so}(3)$.

117 The three $SO(3)$ classes are separated by the Pfaffian of unit elements $(0, +\frac{1}{2}, -\frac{1}{2})$ and by the
 118 participation ratio (2 vs. 4). Diagonal and chiral spans are at subspace distance exactly $\frac{1}{2}$ (principal
 119 angles of 45°). An optimization that does not use the classification confirms it: minimizing a span-
 120 only closure residual from 200 random 3-dimensional subspaces of $\mathfrak{so}(4)$ converged to 192 diagonal
 121 and 8 chiral spans, and to nothing else.

122 **Endpoint ambiguity: why the objective can prefer a wrong closed algebra.** BracketNet infers
 123 each action from the two endpoint codes, $a = q(z, z')$. For the true $SO(3)$ action this is ill-posed.
 124 A rotation about the axis through z fixes z , so g and $g \exp(t\hat{z})$ have the same endpoints. A chiral
 125 $\mathfrak{su}(2)$ has no such ambiguity: left multiplication by unit quaternions acts simply transitively on every
 126 sphere of $\mathbb{R}^4 \cong \mathbb{H}$.

127 **Proposition 3** (Endpoint ambiguity). *On the $SO(3)$ benchmark, use the ground-truth state as the*
 128 *latent code. (a) With chiral generators, the endpoint rule $q(z, z') = \log(z' z^{-1})$ makes transport and*
 129 *composition exact on every triple whose coefficients lie in the coefficient domain. The closure, Gram*
 130 *and covariance terms vanish as well. (b) With the true diagonal generators, no continuous endpoint*
 131 *rule can make transport and composition exact on all triples with pairwise distance $< \epsilon$ on a sphere,*
 132 *for any $\epsilon > 0$. (c) For any action-inference rule based on endpoints, $\mathbb{E} \|\rho(q(z, gz)) - g\|^2 > 0$ under*
 133 *the true action: the stabilizer component of g is not identifiable from (z, gz) .*

134 Part (b) follows from the hairy-ball theorem: exact composition on small triangles would define a
 135 continuous section of $SO(3) \rightarrow S^2$. On the actual data, the minimal-rotation rule for the true algebra
 136 transports exactly but leaves a composition residual of 3.2×10^{-3} ; the chiral rule leaves $\sim 10^{-31}$.

137 **Corollary 1** (Agreement without correctness). *Two models with identical, ground-truth embeddings*
 138 *and $\mathfrak{su}(2)_L$ generators have all of the following:*

- 139 • linear CKA 1 with each other and with the true state;
- 140 • zero closure, transport and composition loss (up to coefficient clipping);
- 141 • cross-model algebra distance 0, but distance $\frac{1}{2}$ to the true algebra;
- 142 • inferred actions that are not the true rotations.

143 **The implication chain, made precise.** Write (i) closure, (ii) correct conjugacy class, (iii) recovery
 144 after latent alignment, (iv) correct action inference.

- 145 • (i) $\Rightarrow$ (ii) holds for T^2 (Prop. 2a) and fails for $SO(3)$ (Prop. 2b).
- 146 • (ii) $\Rightarrow$ (iii) additionally needs an encoder close to an isometry of the state space.
- 147 • (iii) $\Rightarrow$ (iv) fails for $SO(3)$ for *every* endpoint method (Prop. 3c).
- 148 • Agreement between models implies none of (ii)–(iv) (Cor. 1).

149 Proposition 3 describes the loss landscape at one embedding. It does not say which solution gradient
 150 descent reaches; Sec. 6.2 tests that empirically.

151 5 Experimental protocol

152 **Benchmarks.** $SO(2)$, T^2 and $SO(3)$ act linearly on a latent state in $\mathbb{R}^d$ that includes an invariant
 153 coordinate ($d = 3, 5, 4$). The true algebra is therefore a proper subspace of $\mathfrak{so}(d)$, and random spans
 154 do not close: the mean closure of random normalized spans is 0.60 ($K=2$ in $\mathfrak{so}(5)$) and 0.66 ($K=3$
 155 in $\mathfrak{so}(4)$).

156 States are drawn as $z_0 \sim \mathcal{N}(0, I)$, and transitions apply $\exp(\sum_k a_k T_k)$ with $a_k \sim \mathcal{N}(0, 0.24^2)$.
 157 Observations are $x = Qz + 0.18 \tanh(Bz)$, one fixed injective map per group. The rendered

Table 1: Fresh-seed results (seeds 100–119, mean $\pm$ s.d.; frozen protocol). GT dist.: ground-truth algebra distance after Procrustes alignment (chance 0.80 for T^2 , 0.50 for SO(3)). CC: correct-closed runs. SO(2), where all methods coincide, is in Appendix Table 4.

Group	Method	1-step MSE	10-step MSE	Closure residual	GT dist.	CC
T^2	Local	0.0080 $\pm$ 0.0076	0.0645 $\pm$ 0.0841	$3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$	0.161 $\pm$ 0.282	14/20
T^2	+Comp.	0.0081 $\pm$ 0.0076	0.0605 $\pm$ 0.0757	$2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$	0.151 $\pm$ 0.274	14/20
T^2	BracketNet	0.0072 $\pm$ 0.0079	0.0539 $\pm$ 0.0792	$2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$	0.129 $\pm$ 0.282	16/20
T^2	BracketNet ($\lambda=0.1$)	0.0082 $\pm$ 0.0078	0.0622 $\pm$ 0.0785	$3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$	0.157 $\pm$ 0.277	14/20
SO(3)	Local	0.0048 $\pm$ 0.0027	0.0325 $\pm$ 0.0312	$6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.479 $\pm$ 0.435	6/20
SO(3)	+Comp.	0.0048 $\pm$ 0.0031	0.0328 $\pm$ 0.0299	$6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.475 $\pm$ 0.433	5/20
SO(3)	BracketNet	0.0049 $\pm$ 0.0034	0.0319 $\pm$ 0.0341	$2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$	0.243 $\pm$ 0.254	8/20
SO(3)	BracketNet ($\lambda=0.1$)	0.0050 $\pm$ 0.0031	0.0484 $\pm$ 0.0484	$5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$	0.453 $\pm$ 0.413	5/20

benchmark is described in Sec. 6.3. Each run trains on 1,400 triples and is evaluated on 500 ten-edge sequences.

Selection on development seeds only. We fixed the training budget, closure weight and closure objective on development seeds 5–9, with rules committed before those runs:

- *Budget*: the smallest budget at which +COMP. transport is within 10% of its value at 4,000 updates. This rule is method-agnostic.
- *Closure weight*: the λ with the lowest closure among those whose transport is not worse than +COMP.’s. This rule needed an amendment, committed before any fresh run: no λ met the constraint, so the tolerance was relaxed in steps of 10%, and $\lambda = 3$ qualified at 1.20.

The result is 4000 updates, $\lambda = 3$ and Eq. (1). All runs evaluate the final iterate. $\lambda = 0.1$, the originally planned weight, is evaluated at the same budget as a reference.

Seeds and pre-registration. Main evaluation: 20 fresh seeds (100–119). Ablations: 200–209. Rendered benchmark: 300–319. Oracle intervention: 400–419. Each set was used once. Structural thresholds, primary hypotheses and the predictions for the mechanism and transfer experiments were committed before the corresponding runs.

Structural categories (fixed in advance). A run is *collapsed* if $\sigma_{\min}/\sigma_{\max}(B) < 0.1$, and *closed* if its closure is at most 1% of the random-span level. It is *correct-closed* if it is closed and its ground-truth algebra distance (mean squared sine of the principal angles between $\text{span}(RAR^\top)$ and the true span, with R the Procrustes map from codes to true states) is at most 10% of chance. A closed SO(3) span with participation ratio ≥ 3 is *chiral*.

Statistics. Seeds are the independent units; cross-seed quantities use the 10 *disjoint* seed pairs. The primary family compares BracketNet with +COMP., one-sided, with exact paired sign-flip tests and Holm correction: closure (H1), ground-truth algebra distance (H2) and cross-seed algebra distance (H3), each for T^2 and SO(3). Everything else is secondary, and we call a difference detectable only if its 95% interval excludes zero.

6 Results

6.1 Closure makes SO(3) algebras closed, but often wrong

Transition fitting closes T^2 but not SO(3). Without closure regularization, the true T^2 algebra is recovered in 14/20 (+Comp.) and 14/20 (Local) runs, while the SO(3) span stays *non-closed* in 15/20 and 14/20 runs. One- and ten-step transport errors are low in both cases (Table 1). BracketNet at the originally planned $\lambda=0.1$ behaves like the baselines.

Table 2: Pre-registered primary hypotheses, BracketNet vs. +COMP. (one-sided). Units: seeds ($n=20$) for H1–H2, disjoint seed pairs ($n=10$) for H3. Reported: two-sided 95% t -interval of the difference, paired effect size d_z , and the number of units where BracketNet is lower. p : exact sign-flip; Holm over six tests.

Hypothesis	Group	n	+Comp.	BracketNet	95% CI (t)	d_z	lower	p	Holm p
H1 closure	T^2	20	0.2906	0.0216	[-0.5427, +0.0049]	-0.46	16/20	4.3e-05	2.1e-04
H2 ground-truth distance	T^2	20	0.1506	0.1289	[-0.0674, +0.0240]	-0.22	16/20	1.6e-01	3.1e-01
H3 cross-seed distance	T^2	10	0.2138	0.1981	[-0.1421, +0.1106]	-0.09	7/10	3.8e-01	3.8e-01
H1 closure	SO(3)	20	0.6163	0.0250	[-0.8550, -0.3278]	-1.05	20/20	9.5e-07	5.7e-06
H2 ground-truth distance	SO(3)	20	0.4754	0.2429	[-0.3348, -0.1301]	-1.06	19/20	1.2e-04	4.9e-04
H3 cross-seed distance	SO(3)	10	0.6845	0.3922	[-0.4424, -0.1423]	-1.39	9/10	2.9e-03	8.8e-03

189 **Closure regularization: large effects for SO(3), concentrated effects for T^2 .** For SO(3), closure
190 falls from 0.6163 to 0.0250 (lower on 20/20 seeds), and the ground-truth distance falls from 0.4754
191 to 0.2429 (19/20). Both are significant after Holm correction (Table 2).

192 For T^2 the closure test is also significant, but the effect is concentrated: the median paired difference
193 is -8.9×10^{-5} , and the differences are heavily skewed (skewness -1.94). The t -interval includes
194 zero, while the interval obtained by inverting the primary sign-flip test, $[-0.5331, -0.0067]$, does
195 not. The difference comes from the few seeds in which +COMP. fails to close. The T^2 ground-truth
196 distance does not change significantly.

197 Ten-step transport shows no detectable difference in either group; the 95% intervals are
198 $[-0.0199, 0.0066]$ and $[-0.0080, 0.0061]$.

199 **Three SO(3) outcomes.** Every run falls into one of the classes of Prop. 2 or is non-closed (Fig. 1a).

- 200 • BracketNet: 8 correct (diagonal), 8 chiral, 4 non-closed.
- 201 • +COMP.: 5 correct, 0 chiral, 15 non-closed.

202 Closure converts non-closed runs into closed ones, and about half of them land in the wrong class.
203 The rise in correct-closed runs is not significant (exact McNemar $p = 0.25$).

204 6.2 Mechanism: the endpoint objective favours chiral solutions

205 We tested Prop. 3 with predictions registered before any of the following runs.

206 **Composition error by class (existing runs).** As predicted, chiral runs compose far better than
207 correct ones: median composition error 3.3×10^{-5} (chiral) vs. 7.9×10^{-4} (correct). Their one-step
208 errors are the same (0.0035 vs. 0.0034). Their action disagreement with the truth is 1.01 vs. 0.38.
209 The correct runs' value is the floor imposed by Prop. 3(c); chiral runs are no better than the identity.

210 **Closure geometry alone favours the correct class.** Pure closure flow from random subspaces
211 reaches the diagonal class in 192/200 starts (Sec. 4), so the high chiral rate in training is not a property
212 of the closure penalty alone.

213 **Trace of the generator span.** We re-ran the 20 BracketNet SO(3) runs with generator snapshots;
214 20/20 reproduced the stored runs bit-for-bit. We measured each span's chirality $\chi = e_L - e_R$, the
215 share of its energy in $\mathfrak{su}(2)_L$ minus that in $\mathfrak{su}(2)_R$.

- 216 • *Pre-registered.* At the start of the closure ramp, $|\chi|$ predicts a chiral outcome among the 16
217 closed runs with AUC 0.75 (at initialization: 0.61). Medians are 0.22 (ending chiral) vs.
218 0.13 (ending correct).
- 219 • *Post hoc.* The closure residual at the ramp start separates outcomes more sharply (AUC
220 0.94). 7 of the 8 runs that end correct had already nearly closed onto the diagonal during the
221 transition-fitting phase. 7 of the 8 chiral runs were still far from closed when the penalty
222 switched on (Fig. 1b).

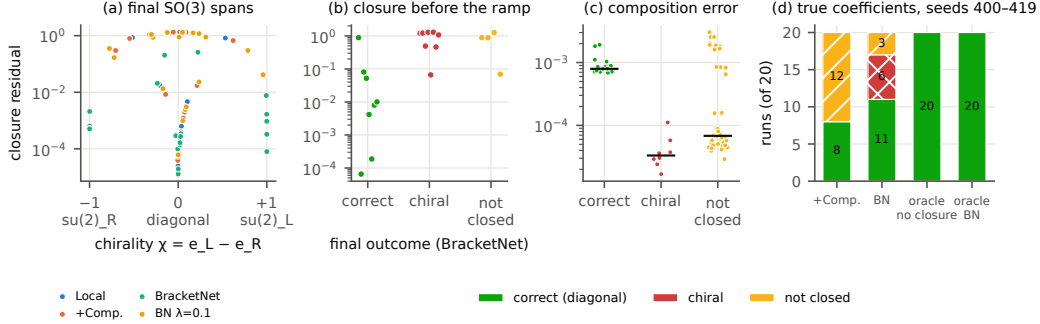


Figure 1: (a) Final SO(3) span chirality χ vs. closure residual for every fresh-seed run: closed runs sit at $\chi \in \{-1, 0, 1\}$, the classes of Prop. 2. (b) Closure of the BracketNet span when the closure penalty switches on, by final outcome (post hoc). (c) Composition error by outcome class, all SO(3) runs. (d) Outcomes with endpoint-inferred vs. true (oracle) action coefficients, seeds 400–419.

Together with the closure flow, this suggests the following picture. Transition fitting sometimes finds the diagonal algebra by itself, and closure then keeps it. When the span is still far from closed, closing it under the transport and composition losses, which favour chiral solutions at the ground-truth embedding (Prop. 3), usually yields a chiral algebra.

Intervention: supplying the true action coefficients (pre-registered). Prop. 3 predicts that chiral solutions lose their advantage if actions need not be inferred from endpoints. On 20 new seeds (400–419) we trained BracketNet with the true coefficients in the transport terms. This is the action-supervised setting of homomorphism autoencoders [Keurti et al., 2023]; q is fit only for evaluation. Results, relative to standard BracketNet on the same seeds:

- chiral outcomes fell from 6/20 to 0/20 (exact McNemar $p = 0.03$);
- correct-closed runs rose from 11/20 to 20/20 ($p = 3.9 \times 10^{-3}$);
- the ground-truth distance changed by -0.215 ($[-0.332, -0.098]$);
- without any closure term, the oracle model also reaches 20/20 correct-closed runs.

Once actions are given, closure regularization is unnecessary: the difficulty lies in inferring actions from endpoints. The oracle models’ own q , which must infer actions from endpoints at test time, still has action disagreement 0.41, the floor of Prop. 3(c). The intervention adds information as well as removing the ambiguity, so it supports the mechanism but does not isolate it.

6.3 A rendered-image benchmark

To test a harder observation process, we rendered the SO(3) state as 20×20 images: the 3-D position of a Gaussian blob sets its image position and size, and the invariant coordinate sets its brightness ($p = 400$). The frozen protocol was applied unchanged on 20 new seeds (300–319), with predictions registered in advance.

- No baseline run closes* (Local 20/20, +COMP. 20/20 non-closed). BracketNet lowers closure on 20/20 seeds (0.062 vs. 1.227, one-sided $p = 9.5 \times 10^{-7}$).
- Every closed BracketNet run is chiral* (7/20; none diagonal). This is read off the chirality invariant, which needs no latent alignment.
- Ground-truth recovery cannot be assessed by our alignment metric here.* The learned latents are far from an isometry of the true state (CKA with it 0.623, Procrustes residual 0.611).
- There is a small transport cost:* 10-step error rises by 4.5×10^{-4} ($[1.4 \times 10^{-4}, 7.7 \times 10^{-4}]$).

Under rendering, closure regularization therefore produced closed algebras only of the wrong class. The benchmark is still synthetic and does not test natural images.

7 Representation alignment: agreement in embeddings vs. agreement in transformations

Models trained from different seeds can be compared at five levels: embedding geometry, transformation algebra, group representation, inferred action, and ground truth. We compare each pair of models on a shared probe set:

- *alignment*: the orthogonal Procrustes map R from $E_a(x)$ to $E_b(x)$, fitted on probe set A;
- *embedding agreement*: linear CKA;
- *algebra agreement*: the cross-seed algebra distance between $\text{span}(RA^{(a)}R^\top)$ and $\text{span}(A^{(b)})$;
- *transfer*: on a held-out probe set B, apply the action inferred by a in b 's space, $\hat{x}_{t+1} = D_b(R\rho_a R^\top E_b(x_t))$. The *transfer gap* is 0 when this is as good as b 's own action and 1 when it is no better than the identity.

Independent units are 10 disjoint seed pairs, the same for every method (Appendix Table 5).

Algebras agree more, through a few discrete classes. For $\text{SO}(3)$, the cross-seed algebra distance falls from 0.6845 to 0.3922 (lower in 9/10 pairs, Holm $p = 8.8 \times 10^{-3}$). For T^2 it does not change significantly.

The $\text{SO}(3)$ pairs separate by class:

- both models correct: distance 0 (1 pair);
- both in the same *wrong* chiral class: distance 0 (1 pair);
- correct vs. chiral: distance $\approx \frac{1}{2}$, as fixed by Prop. 2 (4 pairs);
- at least one model non-closed: 4 pairs.

All 10 +COMP. pairs involve a non-closed model and spread over $[0, 1]$. Closure thus replaces continuous disagreement with disagreement between a few closed classes, so cross-model agreement does not certify correctness (Cor. 1).

Transfer tracks structural agreement, not correctness. Over all 190 pairs of each method (pooled, descriptive):

- $\text{SO}(3)$ pairs of two correct models transfer almost perfectly: median gap 0.013, 0% failures (gap > 0.5), over 63 pairs.
- Pairs of two models in the same wrong class also transfer: median gap 0.177, 0% failures. Aligning them through the ground truth instead of to each other breaks transfer (median gap 4.151).
- Correct-vs-chiral pairs transfer worse: median gap 0.365, 19% failures.

What makes transformations reusable across models is that both models learned the *same* algebra, whether or not it is correct.

CKA vs. structural distance as predictors of transfer failure (pre-registered; prediction not supported). We pre-registered that the label-free algebra distance would predict transfer failure better than CKA. It did not.

- AUC for T^2 : CKA 1.00 vs. algebra distance 0.98.

- AUC for $\text{SO}(3)$: CKA 0.75 vs. algebra distance 0.62; difference interval $[-0.40, 0.30]$ (seed-level cluster bootstrap).

Most transfer failures come from seed-level encoder failures, which CKA detects. CKA is, however, blind to *class* mismatches: $\text{SO}(3)$ correct-vs-chiral pairs have median CKA 0.989, the same as correct pairs (0.991), yet a far larger transfer gap. Embedding similarity and algebra agreement capture different failure modes, and a structural comparison needs both.

8 Ablations

Ablations use seeds 200–209 with same-seed references (Appendix Table 7).

- *Replication.* +COMP. again has higher $\text{SO}(3)$ closure and ground-truth distance than Bracket-Net (+0.2176, $[0.0786, 0.3567]$).
- *Anti-collapse term.* Essential: at weight 1 the T^2 latent distorts (CKA with the truth 0.73) and correct-closed runs fall to 0/10.
- *No detectable effect:* the fit–close–refit schedule (constant penalty or no freeze), the Gram penalty, and a span-only closure objective. We do not claim the schedule is necessary.
- *Misspecified K .* With $K+1$ no run is correct-closed (0/10 for $\text{SO}(3)$), as Prop. 2(c) requires.

9 Limitations

- **Synthetic scope.** Both benchmarks, including the rendered one, are synthetic, low-dimensional and compact. We make no claim about natural images, physical systems or large models.
- **Known K , orthogonal actions.** Closure cannot discover the dimension and presupposes compact (orthogonal) actions.
- **No identifiability.** Wrong closed algebras are frequent for $\text{SO}(3)$, and correct algebras still have irreducible action ambiguity (Prop. 3c). Our ground-truth comparisons use orthogonal Procrustes, so they assume a near-isometric encoder; the T^2 failures are seed-level fitting failures shared by all methods.
- **Limited baselines.** We compare controlled objective variants and an action-supervised oracle; we do not reimplement published methods with different supervision.
- **Statistics and protocol.** We use 20 seeds; effects outside $\text{SO}(3)$ are small or concentrated in few seeds. No transport improvement is established; on rendered images transport is slightly worse. The closure-weight rule needed an amendment, committed before any fresh run. One pre-registered prediction (structure beats CKA at predicting transfer failure) was not supported.
- **Theory is local or benchmark-specific.** The escape bound covers only small actions, the classification is specific to $\mathfrak{so}(4)$ and $\mathfrak{so}(5)$, and Prop. 3 characterizes the loss at the ground-truth embedding, not the optimization path.

10 Conclusion

Penalizing Lie closure changes *how* independently trained transformation models disagree. Continuous scatter over generator spans becomes a choice among a few closed subalgebras. For $\text{SO}(3)$ this raises algebraic agreement between models, and it lets actions transfer between models that made the same choice. But the choice is often wrong, for a reason we can state precisely: actions inferred from endpoints cannot see rotations about the axis through a point, and a chiral algebra hides that blind spot by fitting transitions and compositions exactly. For representation alignment, the lesson is that agreement must be measured at the level of transformations as well as embeddings, and that agreement at either level is not evidence of correctness.

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383 A Proofs

384 **Invariance law.** Under $B \mapsto BM$, $[A'_i, A'_j] = \sum_{k,l} M_{ki} M_{lj} [A_k, A_l]$, so the commutator matrix
 385 becomes $C(M \otimes M)$, while $P_0(BM) = P_0(B)$. Hence $R \mapsto R(M \otimes M)$, and the bounds
 386 follow from $\sigma(M \otimes M) = \{\sigma_i \sigma_j\}$. For $\varepsilon > 0$, $P_\varepsilon(BM) = BM(M^\top GM + \varepsilon I)^{-1} M^\top B^\top =$
 387 $B(G + \varepsilon M^{-\top} M^{-1})^{-1} B^\top$.

388 **Proposition 1.** (a) The commutator is bilinear and antisymmetric and satisfies Jacobi, so closure
 389 makes $\mathfrak{a}$ a Lie subalgebra; the Lie subgroup theorem [Hall, 2015, Ch. 5] gives the integrating subgroup.
 390 It need not be closed: an irrational line in a torus is an example.

391 (b) Write $X = \sum x_i A_i$, $Y = \sum y_j A_j$ and $r_{ij} = N[A_i, A_j]$. Then $\|N[X, Y]\| \leq$
 392 $\|x\| \|y\| (\sum \|r_{ij}\|^2)^{1/2} = \|x\| \|y\| K \sqrt{\mathcal{L}_{\text{close}}}$ by Cauchy–Schwarz, and $\|X\|^2 = x^\top G x \geq \lambda_{\min} \|x\|^2$.

393 (c) Dynkin’s form [Bonfiglioli and Fulci, 2012] is $\log(e^X e^Y) = \sum_{m \geq 1} \frac{(-1)^{m-1}}{m} \sum \frac{[w]}{|w| \prod r_i! s_i!}$,
 394 summed over words $w = X^{r_1} Y^{s_1} \dots X^{r_m} Y^{s_m}$ with $r_i + s_i \geq 1$, where $[w]$ is right-nested.

- 395 • By submultiplicativity, $\|[w]\| \leq 2^{|w|-1} r^{|w|}$.
- 396 • Let e_n bound $\|N[w]\|$ for $|w| = n$, so $e_1 = 0$ and $e_2 \leq \delta r^2$. Write $[w] = [W, Z]$ with W a
 397 letter.
- 398 • $[W, P_0 Z]$ has both arguments in $\mathfrak{a}$, so its escape is at most $\delta r \|Z\| \leq \delta 2^{n-2} r^n$; and
 399 $\|[W, N Z]\| \leq 2r e_{n-1}$. By induction $e_n \leq (n-1) 2^{n-2} \delta r^n$.
- 400 • Each word therefore contributes at most $\frac{\delta}{4} (2r)^{|w|} / \prod r_i! s_i!$. The full sum is $\sum_m \frac{1}{m} (e^{4r} -$
 401 $1)^m = -\ln(2 - e^{4r})$ for $r < \ln 2/4$.
- 402 • The length-1 words lie in $\mathfrak{a}$ and account for $4r$; subtracting them gives the bound.
- 403 • The length-2 words sum to $\frac{1}{2} [X, Y]$, whose escape is at most $\frac{1}{2} \delta r^2$; replacing their crude
 404 contribution $4\delta r^2$ gives $\frac{1}{2} \delta r^2 + O(\delta r^3)$.
- 405 • The series converges absolutely on this ball, and the principal logarithm is analytic there
 406 $(\|e^X e^Y - I\| < 1)$. Both sides agree near 0, hence on the whole connected ball.

407 (d) With $d_t = \hat{z}_t - z_t$: $d_{t+1} = R_t d_t - e_t$ and $\|R_t d_t\| = \|d_t\|$. $\square$

408 **Proposition 2.** Subalgebras of $\mathfrak{so}(d)$ carry the ad-invariant negative-definite form $\text{tr}(XY)$, hence
 409 are compact and reductive.

410 (a) A 2-dimensional reductive algebra is abelian. It is maximal abelian because $\text{rank } \mathfrak{so}(5) = 2$, and
 411 maximal abelian subalgebras of a compact Lie algebra are conjugate [Adams, 1969].

412 (b) A 3-dimensional reductive subalgebra $\mathfrak{h}$ of $\mathfrak{so}(4)$ cannot be abelian (the rank is 2), so $\mathfrak{h} \cong \mathfrak{su}(2)$.
 413 Its projections onto the ideals $\mathfrak{su}(2)_L, \mathfrak{su}(2)_R$ are zero or injective, because $\mathfrak{h}$ is simple.

- 414 • If one projection is zero, $\mathfrak{h}$ is a factor. The factors are ideals, hence invariant under all of
 415 $SO(4)$, and a reflection exchanges them.
- 416 • Otherwise $\mathfrak{h}$ is the graph of an automorphism of $\mathfrak{su}(2)$. That automorphism is inner, and
 417 $SU(2) \times SU(2) \rightarrow SO(4)$ is onto, so $\mathfrak{h}$ is conjugate to the diagonal. The diagonal is the
 418 stabilizer algebra of a unit vector, i.e. $\mathfrak{so}(3)$ on $\mathbb{R}^3 \oplus \mathbb{R}$.

419 The Pfaffian is $SO(4)$ -invariant and odd under reflections. It vanishes on rank-2 rotations and equals
 420 $\pm \frac{1}{2}$ on unit elements of $\mathfrak{su}(2)_{L/R}$.

421 (c) A 4-dimensional reductive subalgebra is $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with the $\mathfrak{u}(1)$ central. The centralizer of
 422 the diagonal $\mathfrak{so}(3)$ is trivial, because $\mathfrak{so}(4) \cong \Lambda^2 \mathbb{R}^4$ decomposes as $\mathbf{3} \oplus \mathbf{3}$ under it, with no trivial
 423 summand. So the $\mathfrak{su}(2)$ must be a factor, and the span cannot contain the diagonal. $\square$

424 **Proposition 3.** Here $g \in SO(3)$ acts on $z_{1:3}$ and fixes z_4 , so transitions preserve $\|z\|$.

425 (a) Identify $\mathbb{R}^4$ with the quaternions $\mathbb{H}$. The exponential of $\mathfrak{su}(2)_L$ is left multiplication by unit
426 quaternions, which acts simply transitively on each sphere $\|z\| = \rho > 0$. For $\|z\| = \|z'\|$, $u = z'z^{-1}$
427 is the unique unit quaternion with $L_u z = z'$.

- 428 • $q = \log u$ is continuous wherever $u \neq -1$.
- 429 • $u_{02} = z_2 z_0^{-1} = (z_2 z_1^{-1})(z_1 z_0^{-1}) = u_{12} u_{01}$, so composition is exact.
- 430 • Closure is exact, the basis is orthonormal so the Gram penalty is 0, and $\text{Cov}(z) = I$ in
431 population.

432 Only coefficients beyond the bound 0.65 break exactness.

433 (b) Suppose a continuous rule gives $R(z, z') \in SO(3)$ with $R(z, z')z = z'$ and $R(z_0, z_2) =$
434 $R(z_1, z_2)R(z_0, z_1)$ whenever all pairwise distances are $< \epsilon$ on a sphere $S \cong S^2$.

- 435 • Then $R(z, z) = I$, and products of R along ϵ -chains depend only on the endpoints and on
436 the homotopy class of the chain, since homotopic chains differ by small-triangle moves.
- 437 • S is simply connected, so fixing $e \in S$, $s(z) = R_{\text{chain}}(e \rightarrow z)$ is a continuous map
438 $S \rightarrow SO(3)$ with $s(z)e = z$.
- 439 • For a unit $f \perp e$, $z \mapsto s(z)f$ is then a nowhere-vanishing tangent field on S^2 , contradicting
440 the hairy-ball theorem.

441 (c) For fixed endpoints (z, z') , the set $\{g : gz = z'\}$ is a coset of the circle stabilizing z . Under
442 Gaussian coefficients the conditional law of g on that circle has a density, so $\mathbb{E}\|\hat{g} - g\|^2 \geq \mathbb{E} \text{tr Cov}(g |$
443 $z, z') > 0$ for every estimator $\hat{g}$. $\square$

444 **Corollary 1.** This is Prop. 3(a) applied to both models, together with Prop. 2(b) and the distance $\frac{1}{2}$
445 between the diagonal and a chiral span. $\square$

446 B Protocol details

447 **Architecture and optimization.** E, D, q are two-hidden-layer SiLU MLPs of width 64. The latent
448 dimension is d , generators are initialized as $W \sim \mathcal{N}(0, 1/d)$, and coefficients are $0.65 \tanh(\cdot)$.
449 Training uses AdamW (lr 2×10^{-3} , weight decay 10^{-5}), batch 160, gradient clipping at 5, $\varepsilon = 10^{-6}$,
450 Gram weight 1 and anti-collapse weight 16. The anti-collapse weight was chosen in a preliminary
451 round on seeds 5–6 by a method-agnostic criterion. The final iterate is evaluated, with no early
452 stopping. All runs are CPU-only and bit-identical on repeat.

453 Order of operations.

- 454 1. Selection rules committed.
- 455 2. Development grid (280 runs).
- 456 3. Rules applied by script.
- 457 4. Fresh seeds 100–119 and ablation seeds 200–209, each run once.
- 458 5. Review predictions committed.
- 459 6. Trace, oracle and rendered runs.

460 No setting was changed after any fresh-seed result was seen.

461 **Rendered benchmark.** The image has 20×20 pixels on $[-3.2, 3.2]^2$. The blob centre is (z_1, z_2) ,
462 its width is $0.45 e^{0.3z_3}$ and its amplitude is $1 + 0.5 \tanh z_4$. Positions are scaled by 0.7 so that blobs
463 stay in view. Centre, width and amplitude determine z , so the map is injective.

464 **Oracle intervention.** The transport terms use the true coefficients of each training edge. The
 465 composition term is dropped, because a_{02} is a product rather than a single coefficient. q is trained
 466 only by regression to the true coefficients, for evaluation.

467 C Statistical audit

468 The six primary tests were recomputed from the raw run files by an independent implementation, and
 469 all matched. The t -interval can disagree with the primary sign-flip test when differences are skewed,
 470 so we also report the interval obtained by inverting the two-sided exact sign-flip test, which matches
 471 the primary procedure, and a percentile bootstrap interval.

Table 3: Paired differences (BracketNet — +Comp.). Intervals: t ; sign-flip-inverted; bootstrap.

Test	Group	n	median diff.	skew	t 95%	inverted sign-flip 95%	bootstrap 95%
H1 closure	T^2	20	-0.0001	-1.94	[-0.5427, +0.0049]	[-0.5331, -0.0067]	[-0.5410, -0.0380]
H2 GT distance	T^2	20	-0.0000	-2.58	[-0.0674, +0.0240]	[-0.0659, +0.0172]	[-0.0688, +0.0136]
H3 cross-seed	T^2	10	-0.0001	-0.33	[-0.1421, +0.1106]	[-0.1410, +0.1091]	[-0.1217, +0.0862]
H1 closure	SO(3)	20	-0.6501	-0.18	[-0.8550, -0.3278]	[-0.8566, -0.3269]	[-0.8295, -0.3572]
H2 GT distance	SO(3)	20	-0.3321	0.21	[-0.3348, -0.1301]	[-0.3347, -0.1293]	[-0.3231, -0.1378]
H3 cross-seed	SO(3)	10	-0.3466	0.30	[-0.4424, -0.1423]	[-0.4464, -0.1439]	[-0.4137, -0.1664]

472 **Threshold sensitivity (robustness check, not pre-registered).** SO(3) fresh-seed counts (correct
 473 / chiral / other closed / non-closed) under alternative thresholds, as a fraction of the random-span
 474 closure level and of the chance distance:

closed	correct	+Comp.	BracketNet
0.1%	5%	3/0/0/17	8/5/0/7
0.1%	10%	3/0/0/17	8/5/0/7
0.1%	20%	3/0/0/17	8/5/0/7
1%	5%	5/0/0/15	8/8/0/4
1%	10%	5/0/0/15	8/8/0/4
1%	20%	5/0/0/15	8/8/0/4
5%	5%	7/0/0/13	8/9/1/2
5%	10%	7/0/0/13	9/9/0/2
5%	20%	7/0/0/13	9/9/0/2

476 The qualitative conclusions (no chiral solutions without closure; several with it; more correct-closed
 477 runs with closure) hold at every setting.

478 D Comparison of settings

Work	Supervision	Learned structure	Closure	Cross-model structure
Rao & Ruderman; Sohl-Dickstein et al.	transformed pairs	linear generators	not enforced	no
Cohen & Welling 2014	pairs, commutative group	irreps of a torus	abelian by design	no
LieGAN / LaLiGAN	distributional invariance	generators (latent)	not penalized	no
Homomorphism AE	transitions + action labels	homomorphism into matrix group	via composition	no
CKA / shape metrics	trained networks	—	—	embeddings only
This work	transitions, no labels	generators + endpoint actions	closure penalty	algebras, transfer

479 These summaries describe each setting at the level needed to judge overlap; see the cited papers for
 480 details.

481 E Additional results

Table 4: Fresh-seed results for all groups, including SO(2) (closure identically 0 for one generator).

Group	Method	1-step MSE	10-step MSE	Closure residual	GT dist.	CC
SO(2)	Local	0.0026 ± 0.0010	0.0061 ± 0.0019	0 ± 0	0.000 ± 0.000	20/20
SO(2)	+Comp.	0.0026 ± 0.0010	0.0065 ± 0.0028	0 ± 0	0.000 ± 0.000	20/20
SO(2)	BracketNet	0.0026 ± 0.0010	0.0063 ± 0.0027	0 ± 0	0.000 ± 0.000	20/20
SO(2)	BracketNet ($\lambda=0.1$)	0.0026 ± 0.0010	0.0063 ± 0.0027	0 ± 0	0.000 ± 0.000	20/20
T^2	Local	0.0080 ± 0.0076	0.0645 ± 0.0841	$3.4 \times 10^{-1} \pm 7.2 \times 10^{-1}$	0.161 ± 0.282	14/20
T^2	+Comp.	0.0081 ± 0.0076	0.0605 ± 0.0757	$2.9 \times 10^{-1} \pm 6.3 \times 10^{-1}$	0.151 ± 0.274	14/20
T^2	BracketNet	0.0072 ± 0.0079	0.0539 ± 0.0792	$2.2 \times 10^{-2} \pm 5.6 \times 10^{-2}$	0.129 ± 0.282	16/20
T^2	BracketNet ($\lambda=0.1$)	0.0082 ± 0.0078	0.0622 ± 0.0785	$3.1 \times 10^{-1} \pm 6.1 \times 10^{-1}$	0.157 ± 0.277	14/20
SO(3)	Local	0.0048 ± 0.0027	0.0325 ± 0.0312	$6.3 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.479 ± 0.435	6/20
SO(3)	+Comp.	0.0048 ± 0.0031	0.0328 ± 0.0299	$6.2 \times 10^{-1} \pm 5.7 \times 10^{-1}$	0.475 ± 0.433	5/20
SO(3)	BracketNet	0.0049 ± 0.0034	0.0319 ± 0.0341	$2.5 \times 10^{-2} \pm 7.1 \times 10^{-2}$	0.243 ± 0.254	8/20
SO(3)	BracketNet ($\lambda=0.1$)	0.0050 ± 0.0031	0.0484 ± 0.0484	$5.4 \times 10^{-1} \pm 5.4 \times 10^{-1}$	0.453 ± 0.413	5/20

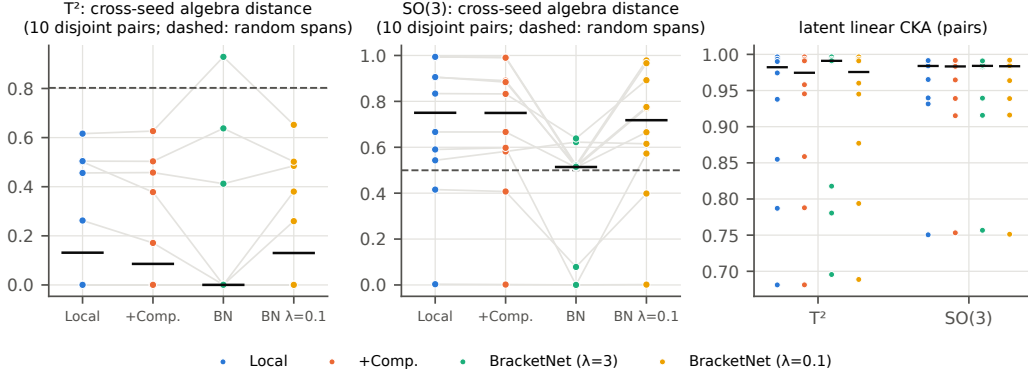


Figure 2: Cross-seed algebra distance on 10 disjoint pairs (grey lines join the same pair across methods; BN = BracketNet) and latent CKA.

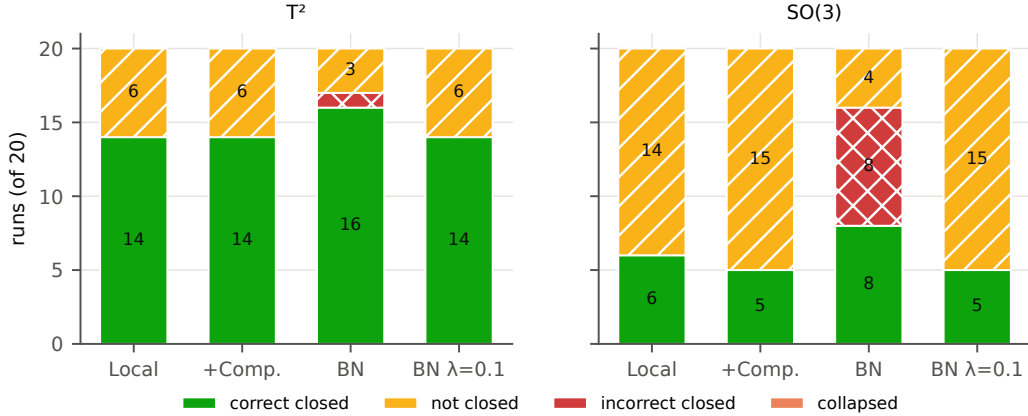


Figure 3: Structural category of each fresh-seed run (BN = BracketNet).

482 F Reproducibility

483 The anonymized supplement contains the data generator, models, training and analysis code, the
 484 selection rules and review predictions, per-run result files for every run, the fresh-seed checkpoints,

Table 5: Cross-seed alignment on 10 disjoint pairs (mean $\pm$ s.d.). Algebra distance chance: 0.80 (T^2), 0.50 (SO(3)). Transfer gap: 0 = native, 1 = identity. Projected: transferred actions re-expressed in b 's generators.

Group	Method	CKA	Algebra dist.	Action disagr.	Transfer gap	Projected gap
T^2	Local	0.920 $\pm$ 0.110	0.234 $\pm$ 0.261	0.596 $\pm$ 0.662	0.876 $\pm$ 1.621	0.647 $\pm$ 1.069
T^2	+Comp.	0.920 $\pm$ 0.109	0.214 $\pm$ 0.252	0.580 $\pm$ 0.659	0.852 $\pm$ 1.573	0.626 $\pm$ 1.032
T^2	BracketNet	0.925 $\pm$ 0.114	0.198 $\pm$ 0.341	0.485 $\pm$ 0.785	0.769 $\pm$ 1.406	0.432 $\pm$ 0.838
T^2	BracketNet ($\lambda=0.1$)	0.923 $\pm$ 0.106	0.228 $\pm$ 0.259	0.592 $\pm$ 0.662	0.770 $\pm$ 1.398	0.611 $\pm$ 0.997
SO(3)	Local	0.951 $\pm$ 0.074	0.685 $\pm$ 0.312	2.280 $\pm$ 0.995	0.745 $\pm$ 1.033	0.923 $\pm$ 0.413
SO(3)	+Comp.	0.949 $\pm$ 0.073	0.685 $\pm$ 0.308	2.222 $\pm$ 0.973	0.741 $\pm$ 1.008	0.909 $\pm$ 0.393
SO(3)	BracketNet	0.952 $\pm$ 0.073	0.392 $\pm$ 0.257	2.015 $\pm$ 0.815	0.732 $\pm$ 0.919	1.257 $\pm$ 0.536
SO(3)	BracketNet ($\lambda=0.1$)	0.949 $\pm$ 0.074	0.664 $\pm$ 0.296	2.194 $\pm$ 0.949	0.726 $\pm$ 0.978	0.924 $\pm$ 0.375

Table 6: Structural breakdown of fresh-seed runs (counts out of 20). Incorrect-closed runs are shown with the chiral count in parentheses. “Bad action” counts correct-closed runs whose action disagreement with the truth exceeds 0.5. PR: participation ratio.

Group	Method	Correct	Incorrect (chiral)	Non-closed	Collapsed	Bad action	Action disagr.	PR
T^2	Local	14	0 (0)	6	0	0	0.336 $\pm$ 0.438	2.98 $\pm$ 0.38
T^2	+Comp.	14	0 (0)	6	0	0	0.326 $\pm$ 0.429	2.93 $\pm$ 0.30
T^2	BracketNet	16	1 (0)	3	0	0	0.261 $\pm$ 0.405	2.86 $\pm$ 0.08
T^2	BracketNet ($\lambda=0.1$)	14	0 (0)	6	0	0	0.331 $\pm$ 0.433	2.93 $\pm$ 0.26
SO(3)	Local	6	0 (0)	14	0	0	0.849 $\pm$ 0.355	2.26 $\pm$ 0.44
SO(3)	+Comp.	5	0 (0)	15	0	0	0.838 $\pm$ 0.361	2.27 $\pm$ 0.44
SO(3)	BracketNet	8	8 (8)	4	0	0	0.764 $\pm$ 0.320	2.91 $\pm$ 1.00
SO(3)	BracketNet ($\lambda=0.1$)	5	0 (0)	15	0	0	0.833 $\pm$ 0.356	2.33 $\pm$ 0.46

figure scripts and unit tests. Every number in the text is generated as a $\LaTeX$ macro from the raw files.

G Use of AI assistance

Code, experiments, analyses and drafts of this manuscript were developed with substantial assistance from an AI coding assistant. The authors are responsible for all content and have checked the claims, proofs and results.

Table 7: Ablations on seeds 200–209 (mean $\pm$ s.d.). Coverage is the fraction of the true span contained in the learned span. Categories: correct / incorrect / non-closed / collapsed. With misspecified K , “correct” is impossible by definition.

Group	Variant	10-step MSE	Closure	GT dist.	Coverage	Categories
T^2	BracketNet (reference)	0.0157 $\pm$ 0.0047	$3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	+Comp. (reference)	0.0231 $\pm$ 0.0275	$2.0 \times 10^{-1} \pm 6.2 \times 10^{-1}$	0.050 $\pm$ 0.156	0.950 $\pm$ 0.156	9/0/1/0
T^2	constant penalty, no staging	0.0370 $\pm$ 0.0475	$1.2 \times 10^{-4} \pm 2.2 \times 10^{-4}$	0.076 $\pm$ 0.170	0.924 $\pm$ 0.170	8/2/0/0
T^2	staged ramp, no freeze	0.0156 $\pm$ 0.0048	$3.4 \times 10^{-5} \pm 2.1 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	no Gram penalty	0.0274 $\pm$ 0.0382	$1.4 \times 10^{-2} \pm 4.5 \times 10^{-2}$	0.027 $\pm$ 0.084	0.973 $\pm$ 0.084	9/0/1/0
T^2	anti-collapse weight 1	0.1997 $\pm$ 0.0253	$3.1 \times 10^{-3} \pm 4.6 \times 10^{-3}$	0.876 $\pm$ 0.130	0.124 $\pm$ 0.130	0/8/2/0
T^2	span-only (whitened) closure	0.0157 $\pm$ 0.0047	$3.6 \times 10^{-5} \pm 2.0 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	0.3 λ	0.0236 $\pm$ 0.0287	$1.4 \times 10^{-1} \pm 4.5 \times 10^{-1}$	0.040 $\pm$ 0.127	0.960 $\pm$ 0.127	9/0/1/0
T^2	3 λ	0.0156 $\pm$ 0.0044	$1.6 \times 10^{-5} \pm 1.1 \times 10^{-5}$	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	10/0/0/0
T^2	$K+1$ generators	0.0248 $\pm$ 0.0194	$3.2 \times 10^{-1} \pm 1.1 \times 10^{-1}$	0.375 $\pm$ 0.103	0.937 $\pm$ 0.154	0/1/9/0
T^2	+Comp., $K+1$	0.0222 $\pm$ 0.0245	$8.2 \times 10^{-1} \pm 1.9 \times 10^{-1}$	0.367 $\pm$ 0.108	0.949 $\pm$ 0.161	0/0/10/0
T^2	$K-1$ generators	0.1917 $\pm$ 0.0300	0 $\pm$ 0	0.501 $\pm$ 0.001	0.499 $\pm$ 0.001	0/10/0/0
SO(3)	BracketNet (reference)	0.0218 $\pm$ 0.0160	$4.2 \times 10^{-2} \pm 8.0 \times 10^{-2}$	0.244 $\pm$ 0.298	0.756 $\pm$ 0.298	5/2/3/0
SO(3)	+Comp. (reference)	0.0393 $\pm$ 0.0309	$7.0 \times 10^{-1} \pm 5.2 \times 10^{-1}$	0.462 $\pm$ 0.412	0.538 $\pm$ 0.412	3/0/7/0
SO(3)	constant penalty, no staging	0.0298 $\pm$ 0.0394	$1.4 \times 10^{-2} \pm 2.6 \times 10^{-2}$	0.185 $\pm$ 0.298	0.815 $\pm$ 0.298	6/1/3/0
SO(3)	staged ramp, no freeze	0.0299 $\pm$ 0.0191	$2.6 \times 10^{-2} \pm 4.6 \times 10^{-2}$	0.239 $\pm$ 0.301	0.761 $\pm$ 0.301	5/2/3/0
SO(3)	no Gram penalty	0.0181 $\pm$ 0.0136	$1.5 \times 10^{-2} \pm 4.3 \times 10^{-2}$	0.177 $\pm$ 0.260	0.823 $\pm$ 0.260	6/2/2/0
SO(3)	anti-collapse weight 1	0.0302 $\pm$ 0.0140	$5.6 \times 10^{-2} \pm 2.2 \times 10^{-2}$	0.619 $\pm$ 0.063	0.381 $\pm$ 0.063	0/1/9/0
SO(3)	span-only (whitened) closure	0.0182 $\pm$ 0.0138	$4.0 \times 10^{-2} \pm 7.7 \times 10^{-2}$	0.243 $\pm$ 0.298	0.757 $\pm$ 0.298	5/2/3/0
SO(3)	0.3 λ	0.0314 $\pm$ 0.0284	$2.2 \times 10^{-1} \pm 3.4 \times 10^{-1}$	0.299 $\pm$ 0.297	0.701 $\pm$ 0.297	4/1/5/0
SO(3)	3 λ	0.0274 $\pm$ 0.0244	$1.7 \times 10^{-2} \pm 3.0 \times 10^{-2}$	0.237 $\pm$ 0.302	0.763 $\pm$ 0.302	5/2/3/0
SO(3)	$K+1$ generators	0.0225 $\pm$ 0.0108	$1.2 \times 10^{-1} \pm 2.4 \times 10^{-1}$	0.457 $\pm$ 0.076	0.723 $\pm$ 0.102	0/7/3/0
SO(3)	+Comp., $K+1$	0.0156 $\pm$ 0.0087	$5.5 \times 10^{-1} \pm 2.6 \times 10^{-1}$	0.511 $\pm$ 0.220	0.652 $\pm$ 0.293	0/0/10/0
SO(3)	$K-1$ generators	0.1253 $\pm$ 0.0535	$6.0 \times 10^{-1} \pm 5.1 \times 10^{-1}$	0.464 $\pm$ 0.166	0.536 $\pm$ 0.166	0/4/6/0

Table 8: Development grid (seeds 5–9; selection only; means over 5 seeds). For each method: 10-step MSE, closure, correct-closed count.

Group	Updates	MSE	Local closure	CC	MSE	+Comp. closure	CC	MSE	BracketNet closure	CC
T^2	420	0.1635	0.1374	1/5	0.1604	0.1104	1/5	0.1647	0.0600	1/5
T^2	1000	0.0704	0.0223	4/5	0.0678	0.0188	4/5	0.0676	0.0011	4/5
T^2	2000	0.0376	0.0016	4/5	0.0391	0.0022	4/5	0.0390	0.0005	4/5
T^2	4000	0.0308	0.0004	4/5	0.0312	0.0003	4/5	0.0317	0.0001	4/5
SO(3)	420	0.1391	0.8088	0/5	0.1532	0.7579	0/5	0.1479	0.3826	0/5
SO(3)	1000	0.0713	0.8762	1/5	0.0843	0.7771	1/5	0.0668	0.2032	1/5
SO(3)	2000	0.0396	0.9245	1/5	0.0308	0.6989	1/5	0.0573	0.1129	2/5
SO(3)	4000	0.0295	0.9574	1/5	0.0276	0.7086	1/5	0.0340	0.0568	2/5

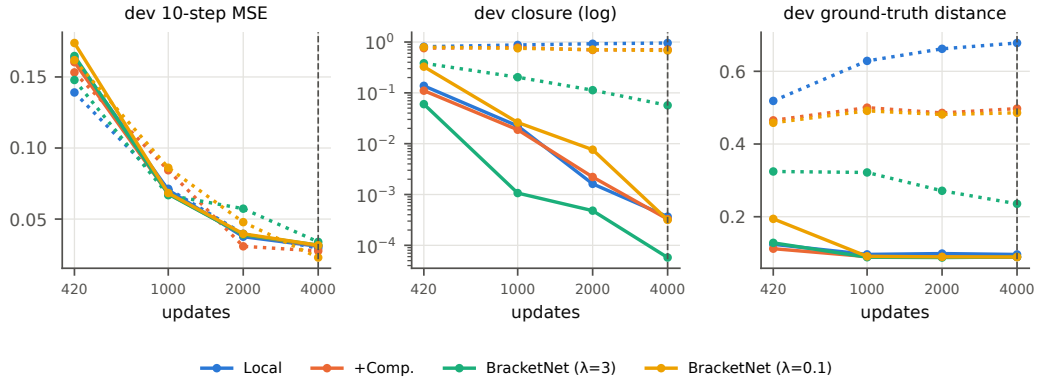


Figure 4: Development-seed budget sweep (solid: T^2 ; dotted: SO(3)); the vertical line marks the selected budget.

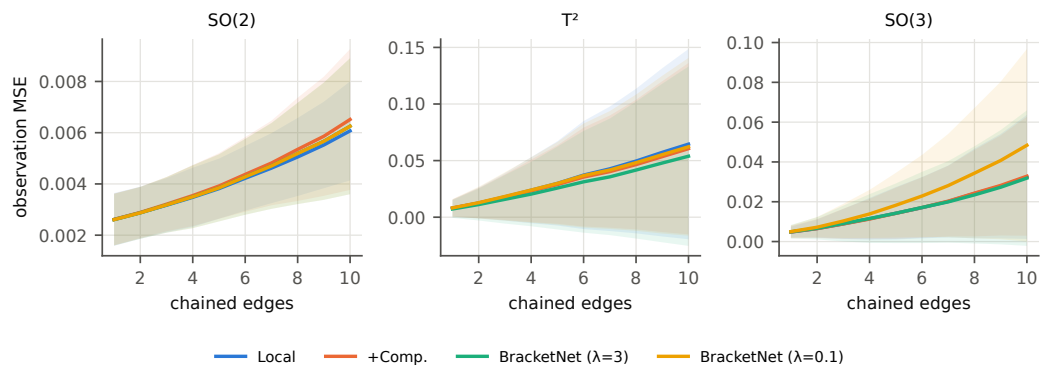


Figure 5: Chained transport error (mean $\pm$ s.d., 20 fresh seeds).